Dan Marshall

AI Engineer and Senior Full-Stack Engineer with 11+ years of software engineering experience. Combines deployed commercial AI systems with hands-on development of DnDan, a pre-alpha roleplaying game engine. This brings a close practical fit to Evoke Interactive's Arkham Catacombs: model integration, reliable application state, backend engineering and AI-driven storytelling.

danmarshall909@gmail.com · github.com/DanMarshall909
linkedin.com/in/developerdan · blog.danmarshall.dev

Relevant skills

AI engineering: AWS Bedrock, OpenAI, Claude API, LangChain4j, retrieval-augmented generation (RAG), guardrails, model selection, local LLMs, prompt engineering and MCP.

Product engineering: C#/.NET, Kotlin/Spring Boot, TypeScript, React, Angular, REST APIs, GraphQL, SQL and event-driven services.

Cloud and delivery: Azure, AWS, Docker, observability, CI/CD, TDD, integration testing, mutation testing and code review.

Recent professional experience

Rapid Circle

Senior Consultant | July 2026 - November 2026

Rewrote CFMEU's wage-claims system from proof of concept to a deployed application employing RAG; fixed security flaws, optimised resource use and added extensive automated tests.

Built screen-activity capture and summarisation with ConnectWise integration, OCR, local LLM support and encryption at rest.

Built an end-to-end proof of concept for Siemens.

Reapit

Senior AI Engineer | September 2025 - February 2026

Workshopped, built and deployed an AI-driven property inspection system using AWS Bedrock, producing structured assessment reports constrained by business rules.

Extended LangChain4j with Bedrock guardrail support, then migrated to the official implementation when upstream support became available.

Selected models against cost, accuracy, features, performance and token usage; delivered with Kotlin, Spring Boot and AWS.

RACT

Senior Software Engineer - Back-end | June 2024 - July 2025

Modernised .NET Framework 4.7.2 components to .NET 9 and built services using Clean Architecture, DDD, MassTransit, WebAPI, GraphQL and SQL Server.

Contributed to an AI coding-rules pilot.

Education

B.Sc. Mathematics and Computer Science - University of South Africa (UNISA).

DnDan: an AI roleplaying game engine

Independent project | Approximately six months of development | Pre-alpha

DnDan aims to capture the freedom of tabletop roleplaying. Areas are intended to be generated progressively as the story unfolds and player choices shape the adventure. Upcoming locations can be prepared ahead of play, while visited places are preserved in the save.

Rules: implementing the full D&D 5e ruleset defined by the System Reference Document (SRD 5.2.1). Full rules coverage is still in development.

Architecture: C# core, provisional Godot client and SQLite saves. AI performs the narrative and characters, while a separate rules engine controls mechanics and world changes.

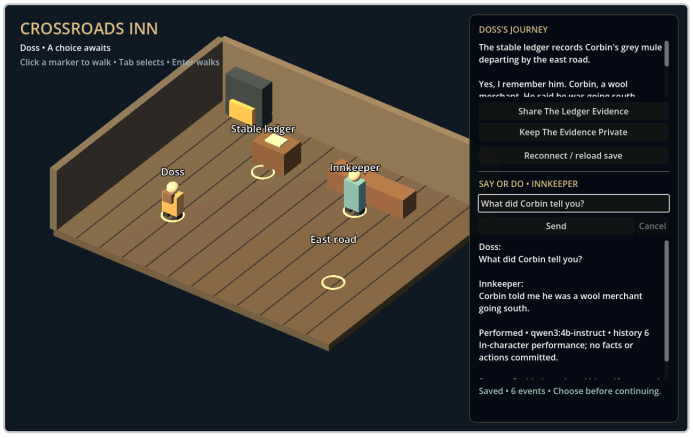
Implemented foundations: exploration and combat with save-and-resume, live character conversations, and bounded preparation of upcoming locations while play continues.

Visual research: initial scene geometry, generated visual treatments and normal/depth-based relighting in Godot.

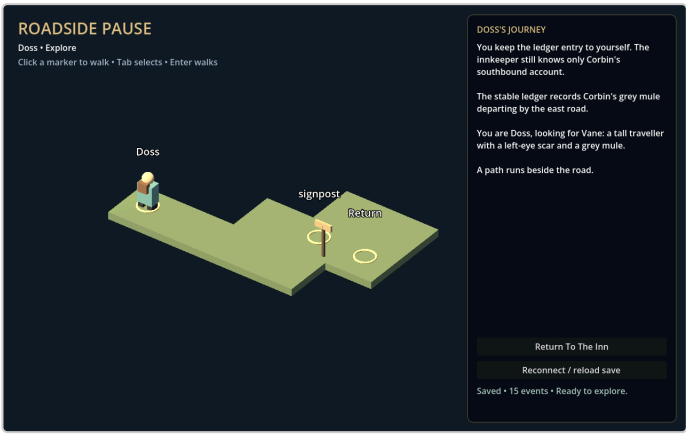
Animation and character equipment: experiments with AI-assisted image-to-3D characters, skeletal rigging and a tested 30 fps walk cycle. Separate armour-fitting studies explore attaching generated equipment to an animated body; garment coverage and clipping still need refinement.

These experiments complement my commercial AI engineering experience and give me practical context for Evoke's narrative RPG ambitions. I am also open to discussing whether licensing or adapting parts of DnDan could accelerate development, subject to technical fit and agreed commercial terms.

The engine in play



Live character dialogue. Local-model conversation alongside recorded evidence, player decisions and save controls.



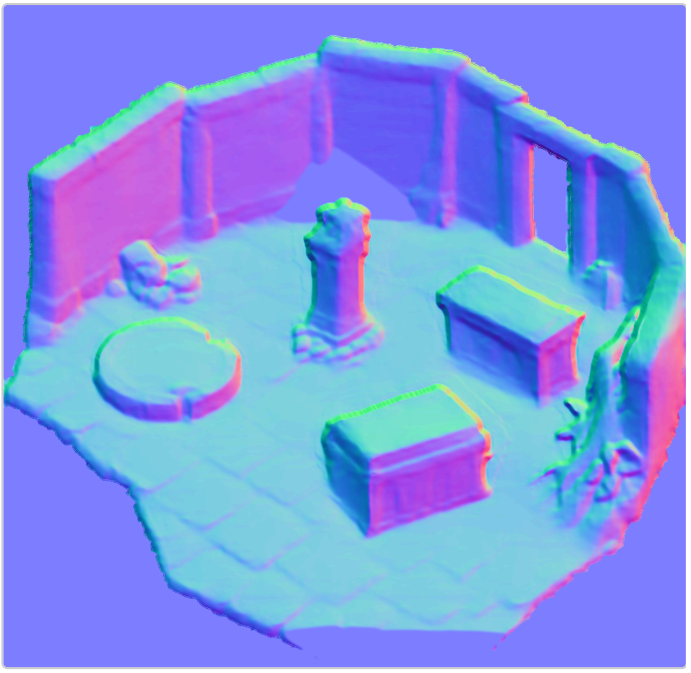
Progressive generation prototype. A bounded, model-authored roadside layout validated by the engine and restored after restart.

Geometry and dynamic lighting

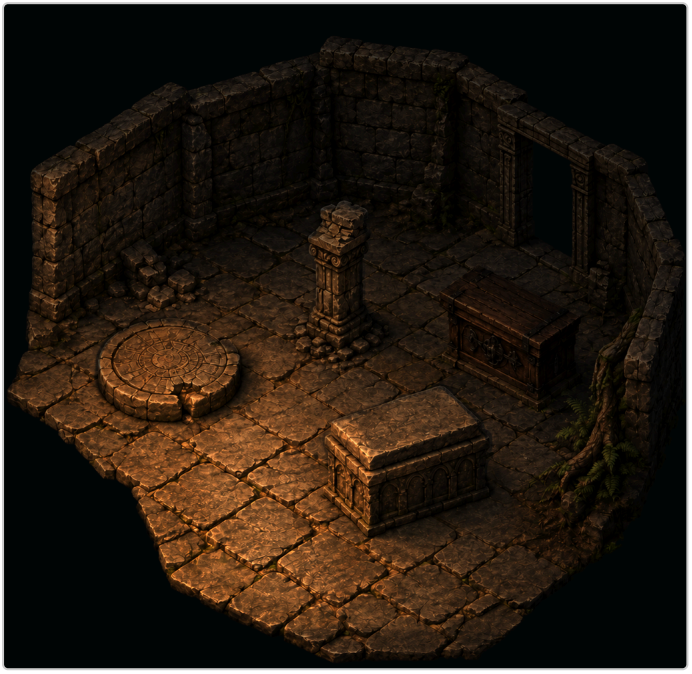
Scene 02 study: initial geometry, reconstructed surface normals and actual Godot relighting captures.



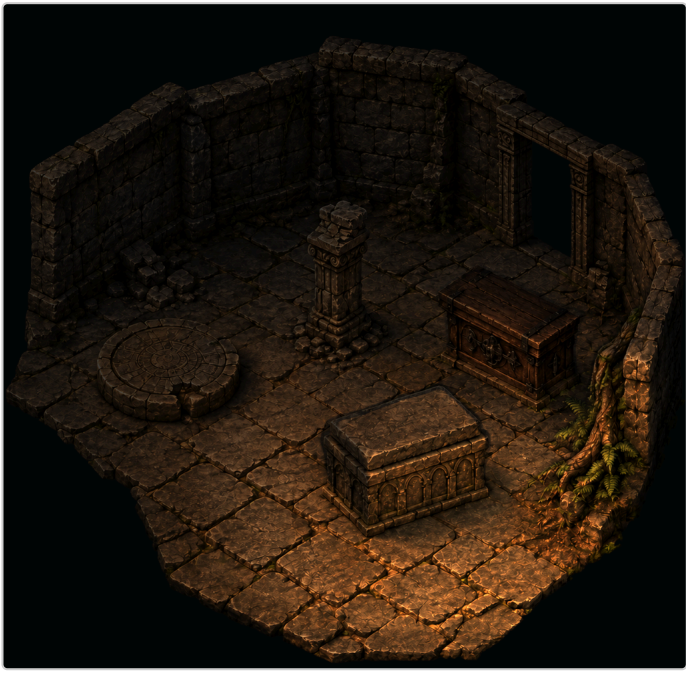
Initial engine geometry. Godot blockout with anchored props and an open doorway.



Presentation normals. Surface normals reconstructed from estimated depth of the generated visual treatment.



Light on the foreground. Godot renderer output using the visual plate, presentation normals and depth.



Alternate light position. Light shifts towards the eastern sarcophagus and roots; the underlying visual plate is unchanged.

The detailed artwork is AI-generated presentation, and its normals are derived from estimated depth. This rendering study is separate from the current gameplay client; matching generated artwork to engine geometry remains under development.

A larger dungeon scene

A multi-room visual study with connected chambers around a central hall.

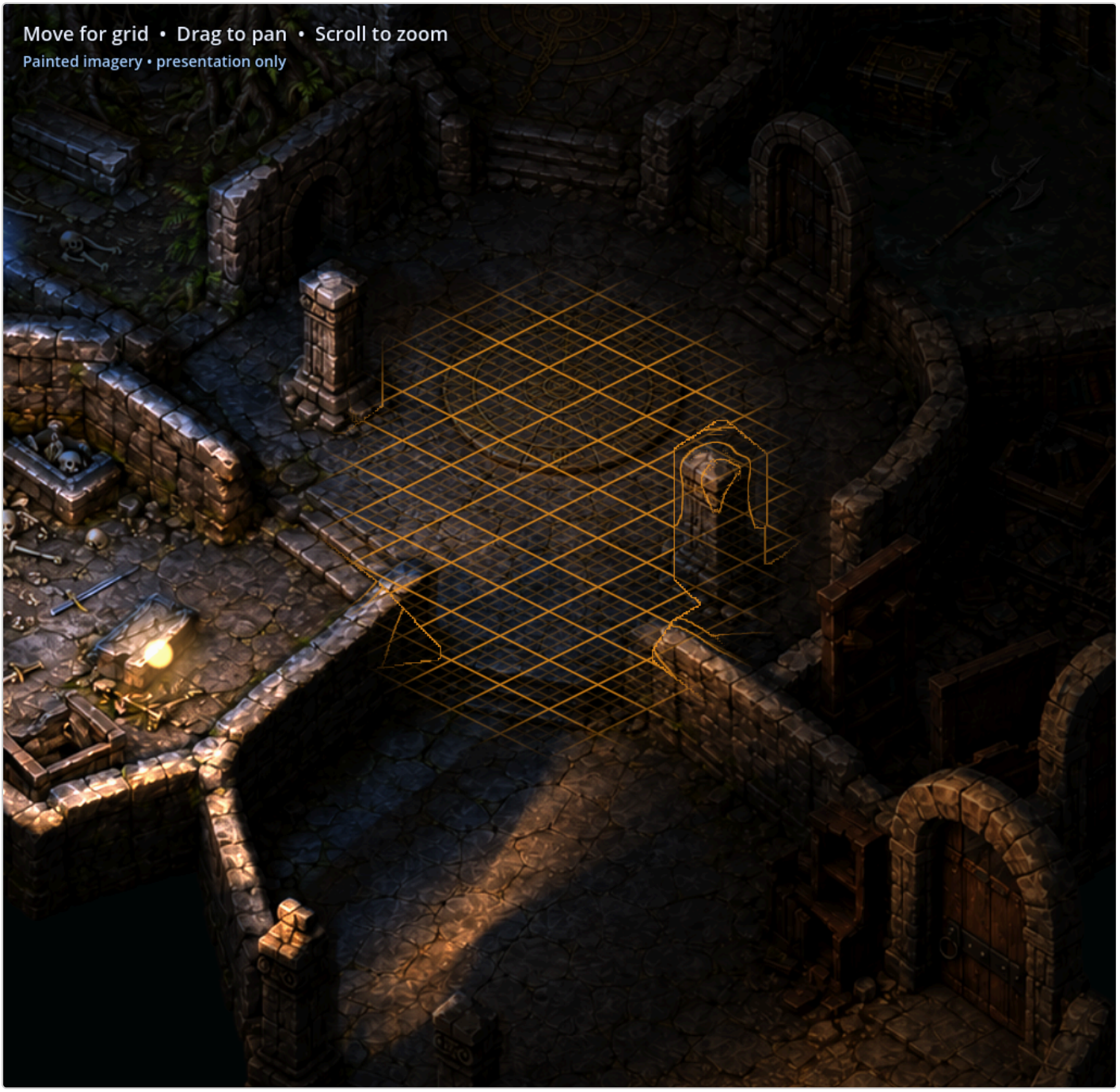


Full dungeon overview. Generated scene artwork shows a crypt, library, flooded chamber and ritual room connected through a central hall.

The following image shows the same dungeon in the Godot renderer, with lighting and a surface grid over the generated floor.

Grid over the generated dungeon floor

Godot renderer capture with a depth-based surface grid over the generated scene.



Grid over generated floor. The surface grid is visible across the central hall and follows depth-derived surfaces around the foreground pillar, while the dungeon artwork remains visible beneath it.

This actual renderer capture shows lighting and the grid overlay together. The grid visualises reconstructed surface depth; it is part of the rendering research in this pre-alpha engine.